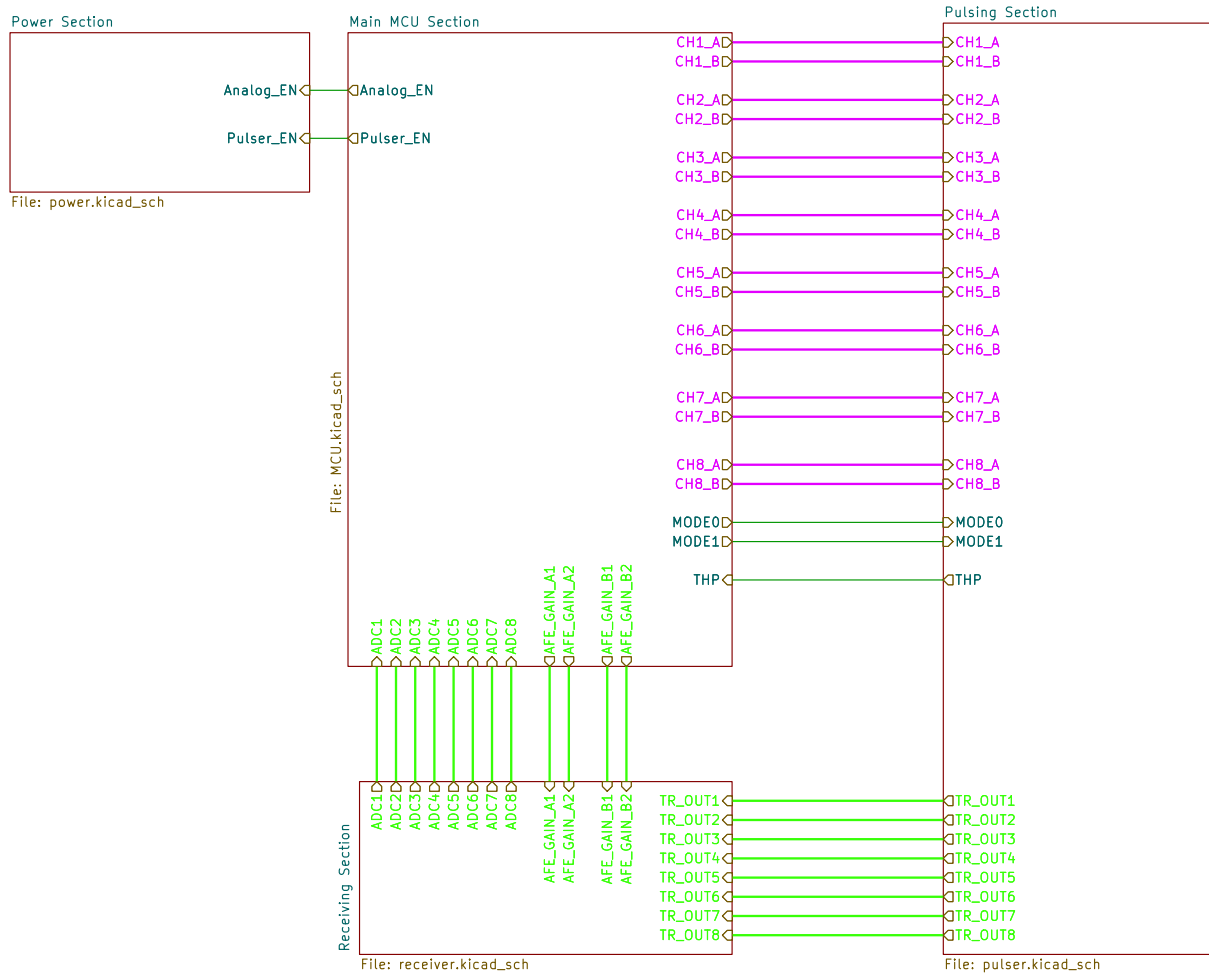
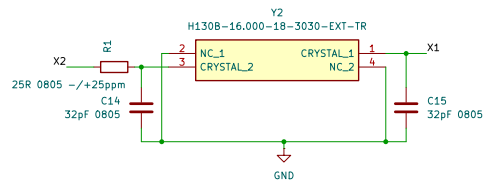
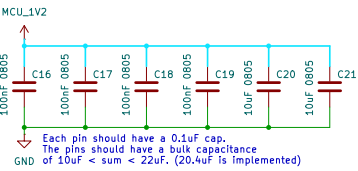
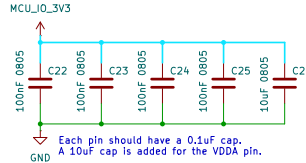




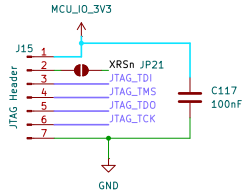
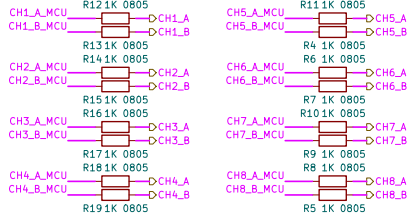
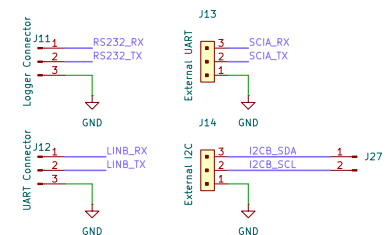
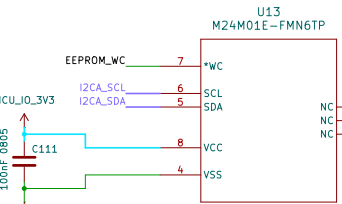
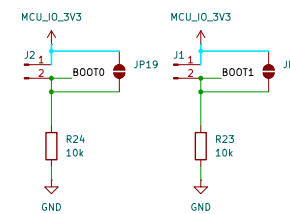
12 Volt (External) (555mA)	7 Volt (Buck Converter) (350mA)	5 Volt (LDO Regulator) (200mA)	LNA + VGA Supply (160mA – short circuit)
			Op-Amps (4 * 10mA = 40mA)
	5 Volt (Buck Converter) (205mA)	3.3 Volt (LDO Regulator) (150mA)	MCU Supply (150mA)
		Positive Pulse Driver Supply (100mA)	
		3.3 Volt (LDO Regulator) (5mA)	Pulser Logic Supply (5mA)
		-5 Volt (Inverting Buck Converter) (100mA)	Negative Pulse Driver Supply (100mA)



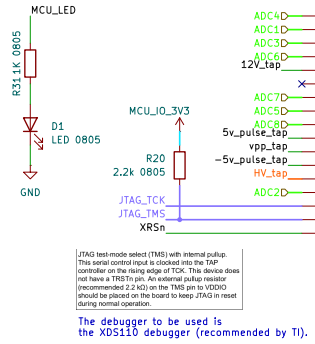
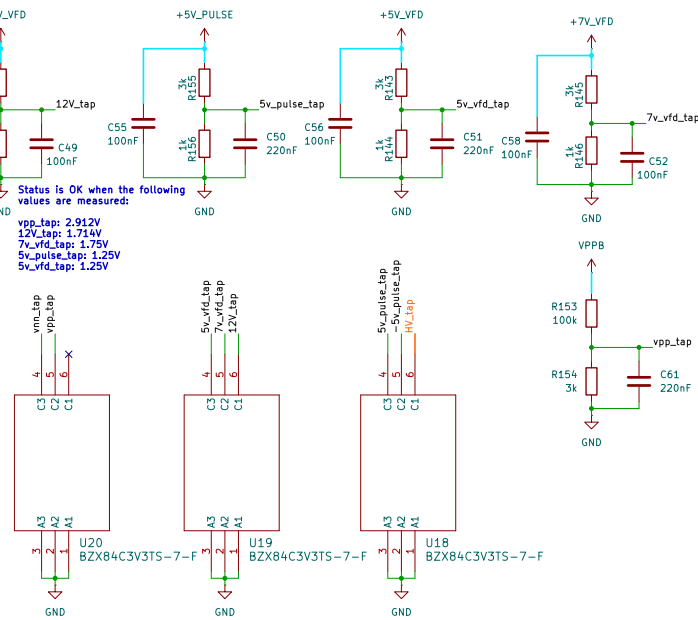
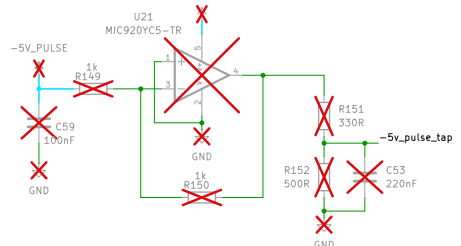
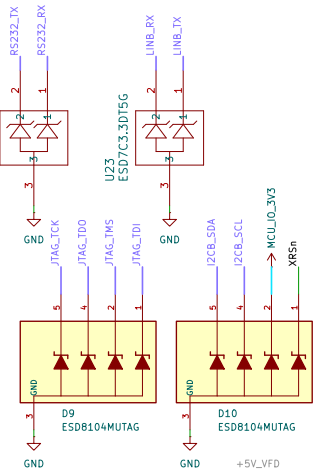
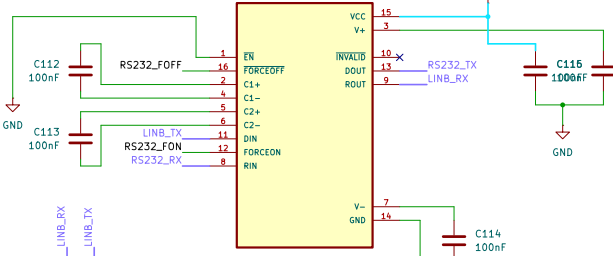
$C1 = C2 = 2(C_{load} - C_{stray}) = 2(18 - 2) = 32pF$
 The crystal is required ($\approx 16MHz$):
 - To have a drive level of $\geq 1mW$ (a damping resistor is used)
 - Have maximum ESR of 75 Ohms (50 is used)
 - $12pF < C1 < 24pF$ (18pF is used)



BOOT MODE	SPROUSE DEFAULT BOOT MODE SELECT	SPROUSE DEFAULT BOOT MODE SELECT
Pin#10	0	0
SPROUSE	0	1
CAN	1	0
Flash	1	1

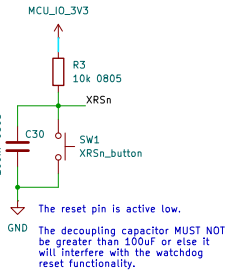


The MCU has no dedicated JTAG reset pin (TRSTn) but theoretically this could work.



JTAG test mode select (TMS) with internal pullup. This serial control input is clocked into the TAP controller on the rising edge of TCK. This device does not have a TRST pin, an external pulser resistor (recommended 2.2 kΩ) on the TMS pin to VDDIO should be placed on the board to keep JTAG in reset during normal operation.

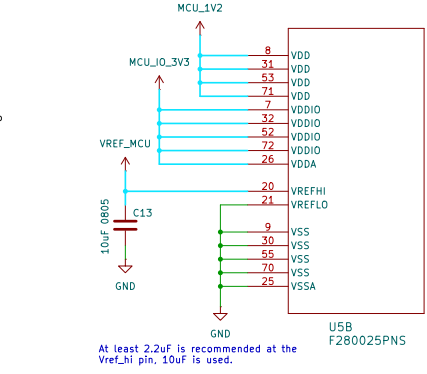
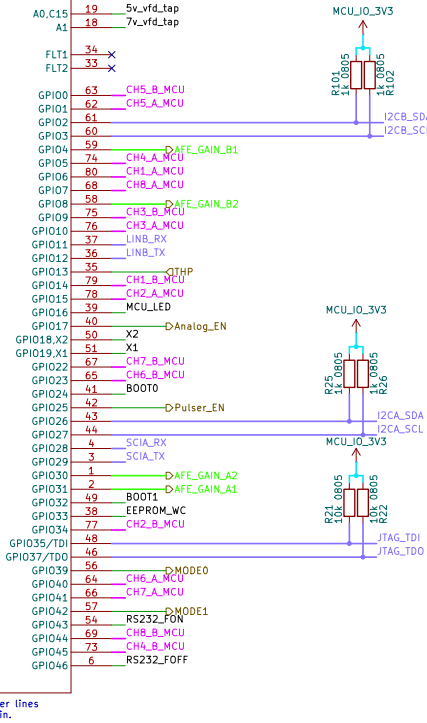
The debugger to be used is the XDS110 debugger (recommended by TI).



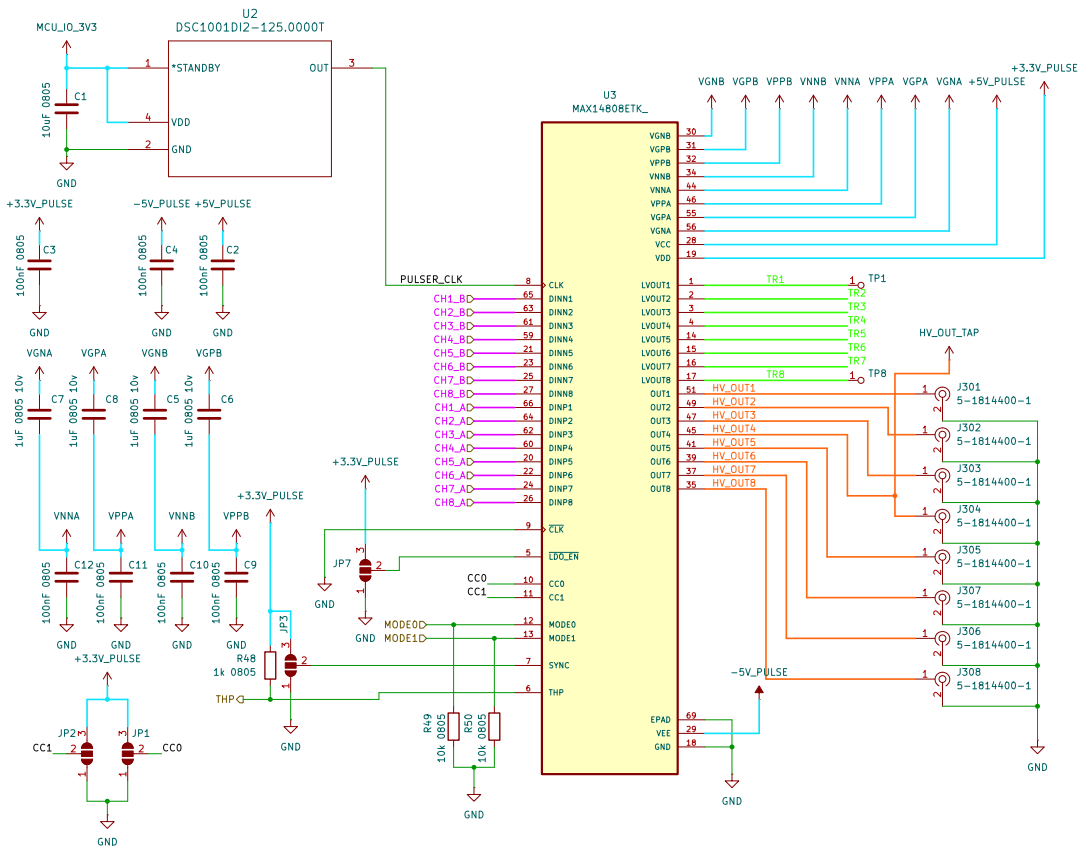
The reset pin is active low.
 The decoupling capacitor MUST NOT be greater than 100uF or else it will interfere with the watchdog reset functionality.



Flash test pins (whatever they are). Must be left unconnected.



At least 2.2uF is recommended at the Vref_hi pin, 10uF is used.

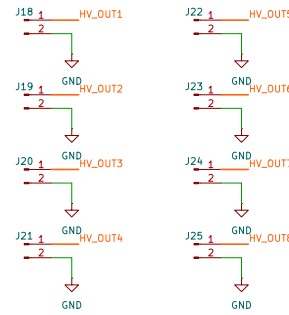


VGXX: gate driver voltage supplies (positive and negative for ports A & B)
VNNX/VPPX: positive and negative HV supplies (for ports A & B)

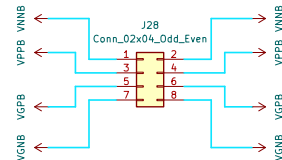
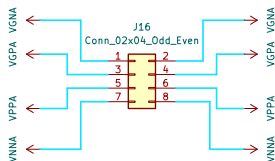
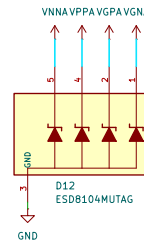
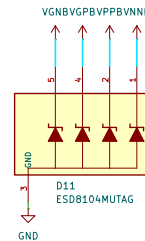
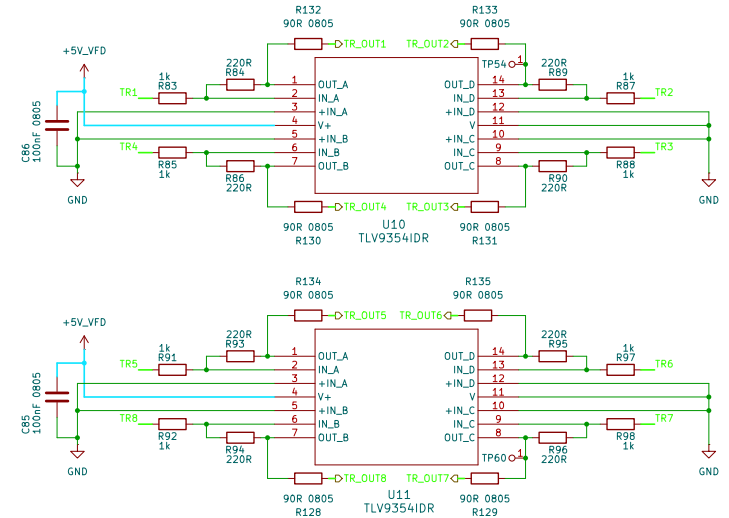
Enabling/Disabling the Internal floating power supplies (FPS):
requirements in most of the cases. Drive LDO_EN low or leave unconnected to enable the internal FPSs; drive LDO_EN high to disable the internal FPSs.
A triple solder jumper is placed to either pull-up (select external FPS) or pull-down (select internal FPS) the LDOEN pin.

Table 5. Current Drive Selection

INPUTS		PULSER OUTPUT CURRENT (typ)
CC0	CC1	
0	0	2A
1	0	1.5A
0	1	1A
1	1	0.5A

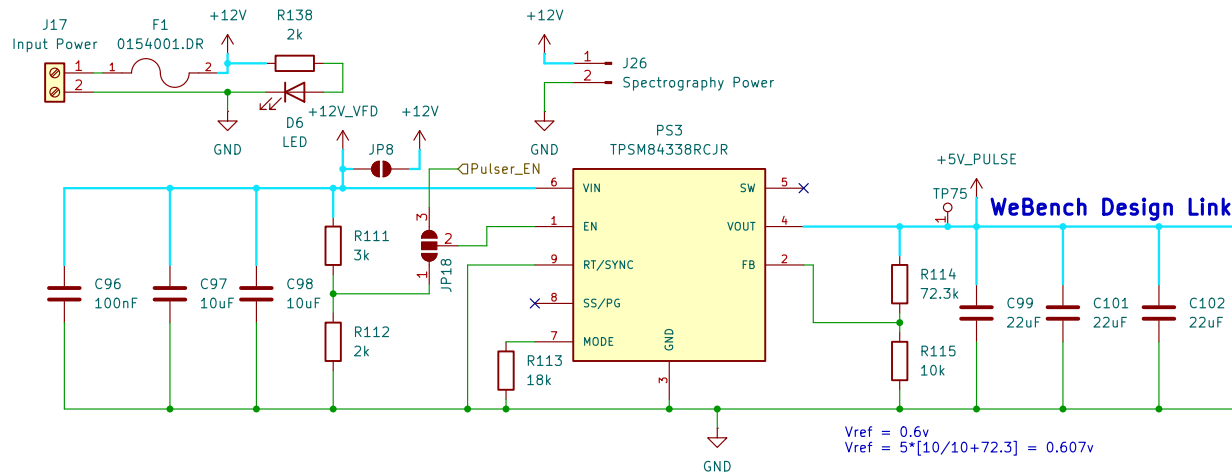
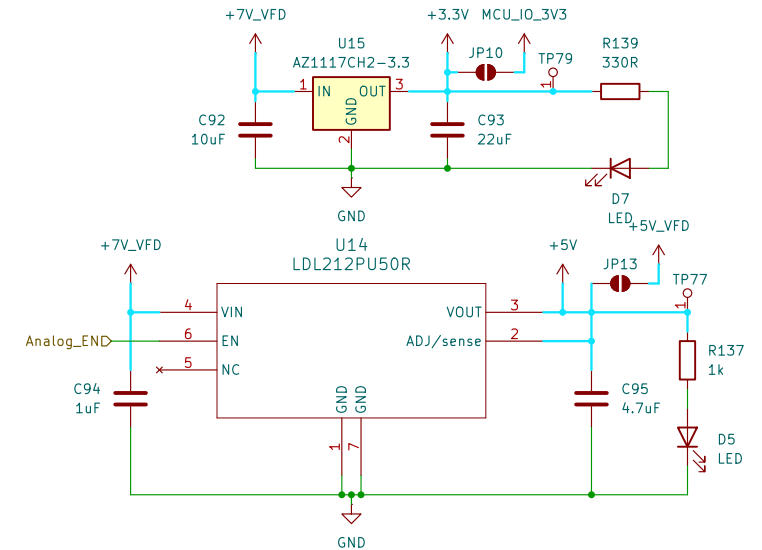
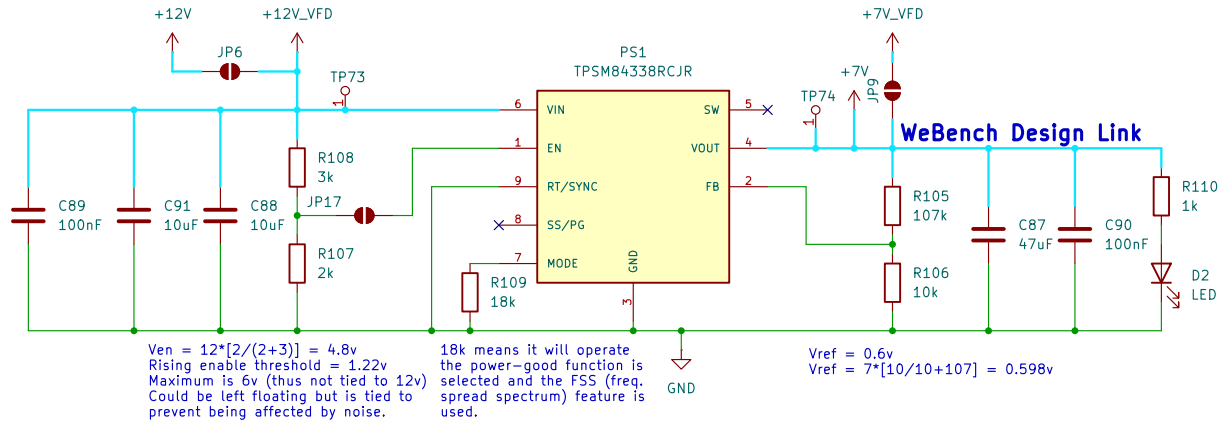


Output swing from the pulser:
-1.2 -> 1.2 (2.4v swing)
Input voltage range for the LNA+VGA:
-0.275 -> 0.275 (0.55v swing)
The differential opamp configuration produces a gain of 0.22 for a swing of 0.528v

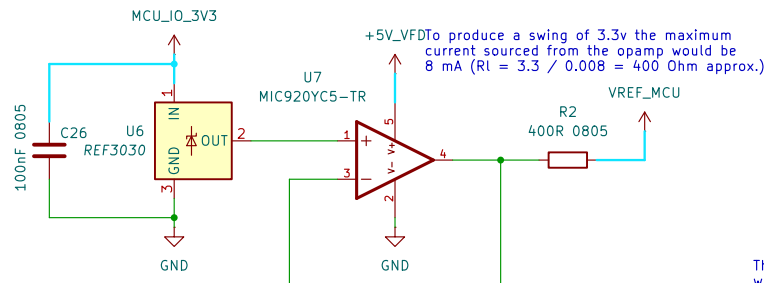
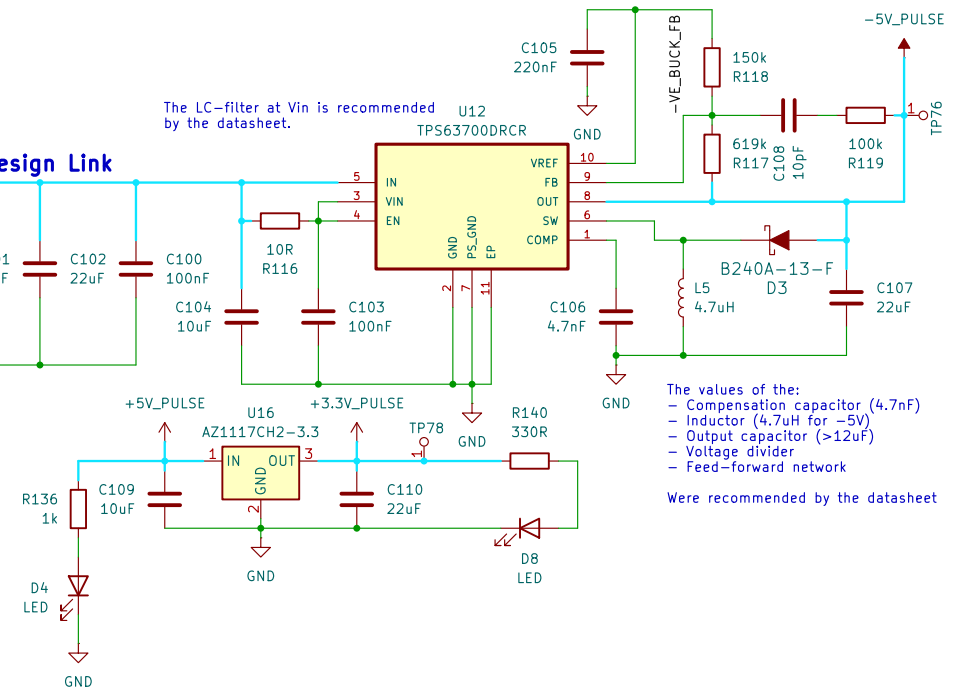


VDD is supplied from an internal 1.2v regulator stepping down from VDDIO but nonetheless its pins need to be decoupled.

In internal VREG mode, tying the VDD pins together is optional as long as each VDD pin has a capacitor connected to pin. See the *VDD Decoupling* section for VDD decoupling configurations.



The LC-filter at Vin is recommended by the datasheet.



These decoupling capacitor values were recommended by the datasheet